Curriculum Vitæ — Bhavin K. Moriya

Research Assistant (**Nov 2023 – Dec 2025**)
Hochschule Esslingen
Flandernstrasse 101
73732 Esslingen
Germany/Deutschland

bhavinmoriya58@gmail.com
Google Scholar, **LinkedIn**
Kaggle Expert, **GitHub**
Ph.D. Thesis YouTube

Latest CV

LONG **SHORT**

Recent Projects

- **Credit Risk Model — PD · LGD · CCF**
- **Vascular Surrogate Models** – Accelerating Intracranial Aneurysm CFD.
- **AudioMNIST** – A spoken digit recognition system.
- **More Projects**

Skills

Programming: Python (Proficient), C++, Rust, SQL, LaTeX, Go, Julia
Data & Simulation: polars, geopandas, SUMO, Scikit-learn, Docker
Workflow: LLM-assisted development (GitHub Copilot), Git, **Agile Research**, **Streamlit**
Languages: English (Fluent), German (**B1/Professional Communication**), Portuguese (Fluent), Hindi (Native), Gujarati (Native)

- Proficient in working with homomorphic encryption. In particular, have used OpenFHE and TFHE-rs to implement custom functions - for example comparison function, polynomial evaluation, square roots etc.
- Proficient in machine learning techniques, including neural networks, recurrent neural networks, bagging and boosting models, regression, and classification.
- Possess a basic understanding of finance principles, including derivative pricing, the Black-Scholes-Merton model, portfolio optimization, CAPM, and Monte Carlo simulation.

Previous Position

A **Data Scientist / Research Assistant** at **Hochschule Esslingen, Germany** during **Nov 2023 – Dec 2025**.

- **AnoMoB Project:** Developed first-of-its-kind encrypted comparison operations and non-linear function evaluations using **OpenFHE (C++)** and **TFHE-rs (Rust)**.
- **Sustainability Analysis:** Leveraged **SUMO** to simulate movement data and conducted comparative analyses of travel modalities focusing on **CO2 emissions** and optimal route duration.
- **Performance Benchmarking:** Co-authored breakthrough research comparing runtime and memory consumption of **CKKS and TFHE** schemes, published in Cryptology ePrint Archive (2025).

- **Data Engineering:** Built production-quality software tools for statistical analysis of mobility data using **Python, polars, and geopandas**.
- **Decision Support for Sustainable Mobility:** Developed a **decision-support tool** for the ANOMOB project that uses a weighted scoring system to help users evaluate the trade-offs between travel time and CO2 emissions. It details a technical architecture based on Dockerized routing engines (ORS/OTP) and highlights research in privacy-preserving mobility analytics and homomorphic encryption.

RESEARCH INTEREST

Mathematical modeling and applications, Statistics, Machine learning, Software developement, and Privacy-enhancing technologies.

TITLE OF THESIS

Some Zero Sum Problems in Combinatorial Number Theory.

PROFESSIONAL EXPERIENCE

- Building a fully homomorphic scheme to get the insight from mobility data (AnoMoB).
- Extensively contributed to the education sector in Brazil since 2015, delivering high-quality instruction.
- Implemented successful projects in machine learning, showcasing expertise in Python, statistics, and mathematics.
- Collaborated extensively and authored 16 research papers published in international journals.
- Post-Doctoral Fellow
  - The Institute of Mathematical Sciences, Chennai
  - August 2, 2011, to August 12, 2013
- National Academy of Sciences Research Associate
  - Harish Chandra Research Institute, Allahabad
  - August 14, 2013, to August 28, 2014
- Visiting Fellow
  - Chennai Mathematical Institute (CMI), Chennai
  - September 7, 2014, to November 6, 2014
- Post-Doctoral Fellow
  - The University of Brasilia, Brasilia, Brazil
  - November 2014 to November 2015
- Assistant Professor
  - The University of Viçosa, Viçosa, Brazil
  - November 2015 to November 2023
- Wissenschaftlicher Mitarbeiter
  - Hochschule Esslingen, Esslingen Am Neckar, Deutschland
  - November 2023 to December 2026.

EDUCATION

- Ph.D. from Harish Chandra Research Institute, Allahabad, spanning from September 24, 2005, to September 28, 2011.
- M. Sc. in Mathematics from The Maharaja Sayajirao University of Baroda, Vadodara in 2005 with 77.05% of marks.
- B. Sc. (***Major :*** Mathematics, ***Subsidiary :*** Physics & Chemistry) from The Maharaja Sayajirao University of Baroda, Vadodara in 2003 with 58.31% of marks (Mathematics with 64.2%).
- Higher Secondary School (Class 12) from Gujarat Board in 2000 with 51.38% marks.
- High School (Class 10) from Gujarat Board in 1998 with 76.14% marks.

PUBLICATIONS

1. (with S. D. Adhikari, M. N. Chintamani and P. Paul) Weighted sums in finite abelian groups. *Unif. Distrib. Theory* 3 (2008), no. 1, 105–110.
2. (with M. N. Chintamani and P. Paul) The number of weighted n-sums. *Int. J. Mod. Math.* 5 (2010), no. 2, 215–222.
3. On zero sum subsequences of restricted size. *Proc. Indian Acad. Sci. (Math. Sci.)* Vol. 120, No. 4, September 2010, pp. 395–402.
4. (with S. D. Adhikari, M. N. Chintamani, Geeta) The Cauchy-Davenport theorem: various proofs and some early generalizations. *Math. Student* 79 (2010), no. 1-4, 109–116.
5. (with M. N. Chintamani) Generalizations of some zero sum theorems. *Proc. Indian Acad. Sci. Math. Sci.* 122 (2012), no. 1, 15–21.
6. (with M. N. Chintamani, W. D. Gao, P. Paul, R. Thangadurai) New upper bounds for the Davenport and for the Erdős-Ginzburg-Ziv constants. *Arch. Math. (Basel)* 98 (2012), no. 2, 133–142.
7. (with C. J. Smyth) Index-Dependent divisors of coefficients of Modular Forms. *Int. J. Number Theory,* Vol. 9, No. 7 (2013), 1841–1853.
8. On Weighted zero sum subsequences of short length. *INTEGERS.* Vol 14, A 21 (2014), 1–8.
9. (with S. Gun) Ramanujan, Quasi-Modular Forms and Transcendence. *Submitted.*
10. (with S. D. Adhikari and Eshita Mazumdar) Relation between two weighted zero-sum constants. INTEGERS, 16, (2016), 1–13.
11. (with A Cambraia Jr, MP Knapp, A Lemos, PHA Rodrigues) On prime factors of Mersenne numbers Palestine Journal of Mathematics Vol. 11(2)(2022), 449–456
12. (with A Lemos, A de Oliveira Moura, A T Silva) On the number of fully weighted zero-sum subsequences, International Journal of Number Theory 15 (05), (2019), 1051-1057.
13. (with A Lemos, BK Moriya, AO Moura, AT Silva) A Variant of Harborth Constant arXiv preprint arXiv:2209.14784

14. (with A Lemos, AO Moura, AT Silva) On the Number of Weighted Zero- sum Subsequences. *Periodica Math. Hung.* Volume 87, pages 366–373, (2023).
15. (with FEB Martínez, A Lemos, S Ribas) The main zero-sum constants over $D_{2n} \times C_2$. SIAM Journal on Discrete Mathematics, volume 37, number 3, pages 1496-1508, 2023.
16. Signed zero sum problems for metacyclic group. Periodica Mathematica Hungarica, pages 1-16, (2024).
17. (with C. Krüger & D. Schoop) A Performance Comparison of the Homomorphic Encryption Schemes CKKS and TFHE, Cryptology ePrint Archive, Paper 2025/1460, 2025.

TALKS AND COLLOQUIUMS

1. Delivered invited talks on Ramsey type theorems at the University of Viçosa, Viçosa, in August 2017 and February 2016, and at the University of Brasilia, Brasilia, in November 2015. Additionally, presented an invited talk on Zero Sum Problems at the University of Brasilia, Brasilia, in May 2015.
2. Conducted tutorials for the course on p-adic numbers at IMSc, Chennai, in 2013.
3. Presented Hensel's Lemma at IMSc, Chennai, on April 16, 2013.
4. Delivered lectures and tutorials during the Summer Program at IMSc, Chennai, held from May 25 to June 20, 2012.
5. Gave an invited talk at Séminaire CAESAR de combinatoire additive at Université Pierre et Marie Curie (Paris 6) on May 31, 2012, titled "Zero-sum subsequences of short length with weights."
6. Presented a Mathematics colloquium at the Institute of Mathematical Sciences, Chennai, on November 24, 2011, titled "Some zero sum problems in combinatorial number theory."
7. Delivered a series of six lectures on Number Theory during the Summer Program in Mathematics (SPIM) 2011 at Harish Chandra Research Institute.

ACADEMIC SERVICE

1. Organized the Summer School at the University of Viçosa, Brazil, from October 19, 2021, to September 20, 2022.
2. Participated in the committee for the Simpósio de Integração Acadêmica - SIA 2016.
3. Organized the Summer School at the University of Viçosa, Brazil, from January 9 to February 22, 2017. Website: `http://www.dma.ufv.br/verao2017/index.php?page=principal`
4. Conducted tutorials on Topology at SPIM 2014, held at Harish Chandra Research Institute from June 16 to July 5, 2014.
5. Conducted tutorials on Zero Sum Problems in AIS Combinatorics, held at Harish Chandra Research Institute from April 28 to May 17, 2014.
6. Conducted tutorials for an Algebra course from December 16 to December 28 at the Annual Foundation School - Part I (2013) held at Kerala School of Mathematics from December 2 to December 28, 2013.

7. Conducted tutorials for the course offered by Dr. K. Sinha on Analytic Number Theory at KIIT University Bhubaneshwar from June 10 to June 30, 2013.
8. Conducted tutorials for the course on p-adic numbers given by Dr. S. Gun in 2013.
9. Co-organized a summer program for M.Sc. students at IMSc with Dr. S. Gun from May 25 to June 20, 2012.

PICME STUDENT (GRADUATES)

- **Luana Costa**
  - *Title*: Aplicação dos Corpos Finitos
  - *Year*: 2019
  - *Institution*: Universidade Federal de Viçosa

MASTER'S THESIS

- **Cynthia Martins Diorio**
  - *Title*: Corpos Finitos e Polinômios de Permutação
  - *Year*: 2019
  - *Institution*: Universidade Federal de Viçosa
- **Luana Souza Marques**
  - *Title*: Método Isoperimétrico em Teoria Aditiva dos Números
  - *Year*: 2018
  - *Institution*: Universidade Federal de Viçosa

PROJECTS AND GRANTS

- **Project Title**: Zero Sum Problems over Finite Abelian Groups
  - **Participants**:
    * Abílio Lemos Cardoso Júnior - Coordinator
    * Allan de Oliveira Moura - Member
    * Anderson Tiago da Silva - Member
    * Bhavinkumar Kishor Sinh Moriya - Member
    * Sávio Ribas - Member
    * Fabio Enrique Brochero Martínez - Member
  - **Financers**: Fundação de Amparo à Pesquisa do Estado de Minas Gerais - Financial Support
- Awarded an ICM Grant by The National Board of Higher Mathematics (NBHM) to attend The International Congress of Mathematicians (ICM 2014) in Seoul, South Korea, from August 13-21, 2014.
- Received a Travel Grant from The National Board of Higher Mathematics (NBHM) to attend The CIMPA-ICTP Research School on Algebraic Curves over Finite Fields and Applications in Manila, Philippines, from July 22 to August 2, 2013.
- Received a Grant to attend the International Congress of Mathematicians from August 19-27, 2010, held in Hyderabad, India.

- Awarded a Grant to attend the School on Combinatorics, Automata, and Number Theory (CANT 2012) at CIRM Marseille, France, from May 21 to May 25, 2012.
- Received a Grant to attend a school on Codes over Rings at METU from August 18 to August 29, 2008, in Ankara, Turkey.
- Received a scholarship from the National Board of Higher Mathematics for Ph.D. studies in 2005.
- Awarded the Dr. M. S. Patel Post Graduate Scholarship on March 11, 2006.

Courses and Activities

1. Conducted a course on Additive Number Theory at the University of Brasilia from August 2015 to December 2015.
2. MAT 140 - Calculus 1, March 2016 - July 2016.
3. MAT 637 - Rings and Modules, March 2016 - July 2016.
4. MAT 147 - Calculus 2, March 2016 - July 2016.
5. MAT 138 - Notions of Linear Algebra, August 2016 - December 2016.
6. MAT 147 - Calculus 2, August 2016 - December 2016.
7. MAT 138 - Notions of Linear Algebra, March 2016 - July 2016.
8. MAT 330 - Algebra 1, August 2017 - December 2017.
9. MAT 138 - Notions of Linear Algebra, August 2017 - December 2017.
10. Conducted a 2-week course in Analysis at the University of Viçosa during the Summer School in February 2017.
11. MAT 209 - Fundamentos da Matemática Elementar III - 2018.
12. MAT 331 - Álgebra II - 2018.
13. MAT 792 - Tópicos Especiais em Matemática III - 2018.
14. MAT 138 - Noções de Álgebra Linear - 2018.
15. MAT 146 - Cálculo I - 2018.
16. MAT 146 - Cálculo I - 2019.
17. MAT 791 - Tópicos Especiais em Matemática II - 2019.
18. MAT 795 - Problemas Especiais II - 2019.
19. MAT 137 - Introdução à Álgebra Linear - 2019.
20. MAT 140 - Cálculo I - 2019.
21. MAT 146 - Cálculo I - 2020.
22. MAT 330 - Álgebra I - 2020.
23. MAT 146 - Cálculo I - 2021.
24. MAT 140 - Cálculo I - 2021.
25. MAT 146 - Cálculo I - 2022.
26. MAT 146 - Cálculo I - 2022.
27. MAT 232 - Matemática Finita - 2022.

28. MAT 146 - Cálculo I - 2023.
29. MAT 232 - Fundamentos de Aritmética - 2023.

CONFERENCES AND WORKSHOPS

1. Participated in the Conference on Anonymization of Integrated and Georeferenced Data 7-8 October, 2024 Spreespeicher in Berlin.
2. Participated in The 24th Privacy Enhancing Technologies Symposium held at Universitz of Bristol, UK from 15–20 July, 2024.
3. Participated in the 12th Privacy Enhancing Techniques Convention: PET-CON 2023 held at TU Berlin (Berlin, Germany) from 13-14 March 2024.
4. Participated in the AnoSiDat Kongress: Anonymisierung für eine sichere Datennutzung held at Media Docks in Lübeck, Germany on 16 & 17 April 2024.
5. Participated in the International Conference in Number Theory and Physics held at IMPA, Rio de Janeiro, Brazil from June 15 to June 26, 2015.
6. Attended The International Congress of Mathematicians 2014 in Seoul, held from August 13 to August 21, 2014.
7. Participated in a School on Algebraic Curves held at Harish Chandra Research Institute, Allahabad from February 25 to March 5, 2014.
8. Attended the CIMPA-ICTP research School on Algebraic Curves over Finite Fields and Applications in Manila, Philippines from July 22 to August 2, 2013.
9. Participated in a course by Prof. J. Oesterlé on "Multiple Zeta Values, the recent advancements" at IMSc from October 5, 2012 to January 10, 2013.
10. Attended the School on Combinatorics, Automata and Number Theory (CANT 2012) at CIRM Marseille, France from May 21 to May 25, 2012.
11. Participated in an "International Meeting on Number Theory Celebrating the 60th Birthday of Professor R. Balasubramanian" at Harish Chandra Research Institute, Allahabad from December 15 to December 20, 2011.
12. Attended a course on "Diophantine approximations and Diophantine equations" by Prof. Michel Waldschmidt at Harish Chandra Research Institute, Allahabad.
13. Participated in a course on "Families of L-functions" by Prof. V. Kumar Murty at IMSc in October 2011.
14. Participated in a course on "Automatic Sequences" by Prof. Jean-Marc Deshouillers at IMSc in October 2011.
15. Attended the RMS 26th Annual Conference held at Allahabad University from October 2 to October 5, 2011.
16. Attended the RMS workshop on Additive Combinatorics held at HRI from September 27 to October 1, 2011.
17. Participated in some courses in a Number Theory Year held at IMSc, Chennai from November 23, 2010 to March 18, 2011.
18. Attended the ICM Satellite Conference on Analytic and Combinatorial Number Theory from August 29 to September 3, 2010 held at IMSc, Chennai.

19. Participated in the International Congress of Mathematicians from August 19 to August 27, 2010 held in Hyderabad, India.
20. Attended the HRI International Conference in Mathematics from March 7 to March 8 and March 16 to March 20, 2009 held in HRI, Allahabad.
21. Participated in the Discussion Meeting on Modular Forms from February 21 to March 6, 2009 held in HRI, Allahabad.
22. Attended the Discussion Meeting on Additive Combinatorics during January 16 to January 31, 2009 held at HRI.
23. Participated in the Codes over Rings workshop at METU from August 18 to August 29, 2008, Ankara, Turkey.
24. Attended the Advanced Instructional School in Algebraic and Analytic Number Theory from December 3 to December 28, 2007 held in HRI, Allahabad.
25. Participated in the International Conference on Number Theory and Cryptography from February 23 to February 27, 2007 held in HRI, Allahabad.
26. Attended the International Conference on Number Theory from December 1 to December 5, 2006 held in HRI, Allahabad.
27. Participated in The Annual Foundation School (AFS)-II from June 1, 2006 to June 28, 2006, jointly organized by Bhaskaracharya Pratishthana, Pune and Dept. of Mathematics, University of Pune.
28. Attended the International Conference on Teichmüller Theory and Moduli Problems, January 2005 held at HRI.

References

- Prof. Dr. Dipl.-Informatiker Dominik Schoop
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  Raum: F 01.354
  Flandernstraße 101
  73732 Esslingen
  Germany
  e-mail: Dominik.Schoop@hs-esslingen.de
- Prof. Jean Paul Allouche
  CNRS, UMR 7586, Institute of Mathematics of Jussieu-PRG
  Combinatorial and Optimization Team
  Sorbonne University
  Case 247
  4 Place Jussieu F-75252 Paris Cedex 05 - FRANCE
  e-mail: jean-paul.allouche@imj-prg.fr
- Dr. Wolfgang Schmid
  LAGA (UMR 7539) et
  Département de Mathématiques et H.S.
  Université Paris 8
  2 rue de la Liberté
  93526 Saint-Denis cedex
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11. Januar 2026
DJS

Arbeitszeugnis für Herrn Bhavinkumar Kishor Sinh Moriya, Ph.D.

Herr Bhavinkumar Kishor Sinh Moriya, Ph.D., geboren am 22.10.1982, war vom 13.11.2023 bis zum 31.12.2025 als wissenschaftlicher Mitarbeiter im vom BMBF geförderten Forschungsprojekt „Anonymisierte Erfassung und Nutzung von Mobilitäts- und Bewegungsdaten (AnoMoB)“ an der Hochschule Esslingen beschäftigt.

Das Forschungsprojekt AnoMoB beschäftigte sich mit statistischen und kryptographischen Methoden, mit denen Trajektorien von Personen im öffentlichen Raum anonym erhoben und verarbeitet werden können.

Als wissenschaftlichem Mitarbeiter des Forschungsprojekts wurden Herrn Moriya die folgenden Aufgaben übertragen:

- Analyse und Bewertung von Verfahren der homomorphen Kryptographie für die Eignung der Implementierung von anonymen Verarbeitungsmethoden von Bewegungsdaten
- Implementierung von Datenverarbeitungsalgorithmen mit Hilfe von Bibliotheken für die homomorphe Kryptographie
- Vergleich von Laufzeiten und Speicherverbrauch von Implementierungen in unterschiedlichen kryptographischen Schemata und Bibliotheken
- Publikation und Präsentation von Forschungsergebnissen auf Konferenzen
- Entwicklung von Softwarewerkzeugen für die Analyse von Bewegungsdaten
- Analyse von simulierten und realen Bewegungsdaten

Herr Moriya hat sich stets sehr gewissenhaft und mit großem Fleiß und Arbeitswillen den gestellten Aufgaben gewidmet. Dabei kamen ihm seine fundierten mathematischen Kenntnisse zu Gute, und er konnte sich ausgesprochen schnell in die mathematischen Konzepte der verschiedenen homomorphen Schemata einarbeiten. Er hat stets eigene sehr gute Lösungsideen selbständig in das Projekt eingebracht.

Bei der Implementierung der Algorithmen mit den Bibliotheken OpenFHE und TFHE-rs hat Herr Moriya seine Kenntnisse in den Programmiersprachen C++, Rust und Go erweitert.

Durch die Entwicklung der Softwarewerkzeuge für die statistische Analyse von Bewegungsdaten hat Herr Moriya seine Kenntnisse in Python vertieft und mit Python-Modulen wie geopandas und polars gearbeitet. Für die Simulation von Bewegungsdaten hat er SUMO genutzt und mit OpenRouteplanner und openrouteservices die in simulierten und echten Bewegungsdaten gewählten Wege und Modalitäten mit den optimalen Wegen bzgl. Reisedauer und CO_2-Ausstoß verglichen.

Herr Moriya hat alle ihm übertragenen Aufgaben stets gründlich und mit großer Sorgfalt bearbeitet und zu meiner vollen Zufriedenheit ausgeführt. Ich habe immer sehr gerne mit ihm zusammengearbeitet.

Sein Verhalten gegenüber Kooperationspartnern, Kollegen und Kolleginnen und seinen Vorgesetzten war stets sehr vorbildlich, freundlich und offen.

Neben seiner fachlichen Arbeit hat Herr Moriya beträchtliche private Zeit und Aufwand investiert, um die deutsche Sprache zu erlernen. Durch seine schnelle Auffassungsgabe und seine Begabung für Sprachen konnte er recht bald nach seiner Einstellung Besprechungen in deutscher Sprache folgen und hat bewusst am Ende seiner Beschäftigung von sich aus die Kommunikation in Deutsch bevorzugt.

Herr Moriya war befristet für die Dauer des Forschungsprojekts AnoMoB angestellt. Sein Vertrag läuft zusammen mit dem Projekt aus. Ein Anschlussprojekt ist nicht bewilligt worden.

Ich danke ihm für seine engagierte Mitarbeit und wünsche ihm für seinen weiteren beruflichen Karriereweg viel Erfolg und alles Gute.

Esslingen, den 11.01.2026

D. Schoop

Prof. Dr. Dominik Schoop